BIO506P – BIOCHEMISTRY II

Practical Exam Notes

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M.C.Qs (Multiple Choice Questions)

1. Number of moles of solute dissolved in $1dm^3$ is called **MOLARITY.**
2. It is used for transfer of precise volume of liquid **BURRETE.**
3. Iodine test is used for the detection of **STARCH.**
4. Yeasts are **FUNGI**.
5. Glycogen is stored in the form of **GLUCOSE** energy.
6. 0.9% NaCl is added in lipid separation in which phase? **PHASE SEPARATION**
7. Accurate sample of urine for glucose test is **5 ml.**
8. The pH of neutral solution is **7.**
9. In the TLC plate, **SILICA GEL** is an adsorbent acts as **STATIONARY PHASE.**
10. Difference between D-Glucose and L-Glucose is the position of **–OH GROUP**.
11. Byproduct of fermentation is **ETHANOL**.
12. Glycogen is stored in liver as **ANIMAL STARCH**.
13. **AMYLOSE** gives **BLUE-BLACK** color with iodine.
14. Industries do fermentation in both **AEROBIC** & **ANEROBIC** conditions.
15. **VOLUMETRIC FLASK** is used to measure exact volume of a liquid.
16. The component of solution that dissolves other component is called **SOLVENT**.
17. Yeast is present in **ONLY FRUITS AND PLANTS**.
18. Adsorbent in TLC plate is **SILICA GEL**.
19. Which of the following is not related to glucose? **INULIN**
20. Structure of amylopectin is composed of **ALPHA 1,4** & **ALPHA 1,6** glycosidic bonds.
21. It can be cause of denaturation of lipids during extraction: **HIGH TEMPERATURE & LIGHT INTENSITY.**
22. Boiling of Fehling's solution can **SPEED UP THE REACTION**.
23. Which term describes the non-moving phase in thin layer chromatography? **STATIONARY PHASE**.

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SHORT QUESTIONS

1. Why pure solvents are used in lipid extraction?

Pure solvents are used in lipid extraction because lipids are non-polar substances and dissolve readily in pure organic solvents such as chloroform, hexane, ether, or petroleum ether. Pure solvents ensure maximum extraction efficiency, prevent contamination from water or impurities, and help obtain accurate results by selectively dissolving lipids while leaving most non-lipid components behind.

2. Differentiate between Amylopectin and Amylose.

Difference Between Amylose and Amylopectin

Amylose	Amylopectin
Linear, unbranched polymer of glucose.	Highly branched polymer of glucose.
Contains α-(1→4) glycosidic bonds only.	Contains α-(1→4) bonds in chains and α-(1→6) bonds at branch points.
Constitutes about 20–30% of starch.	Constitutes about 70–80% of starch.
Less soluble in water.	More soluble in water.
Gives a deep blue/black color with iodine.	Gives a reddish-brown or purple color with iodine.
Has a helical structure.	Has a tree-like branched structure.

3. Define microscopy and its types.

Microscopy

Microscopy is the science and technique of using microscopes to observe objects that are too small to be seen clearly with the naked eye.

Types of Microscopy

1. ***Light (Optical) Microscopy*** – Uses visible light and lenses to magnify specimens.
2. ***Fluorescence Microscopy*** – Uses fluorescent dyes to visualize specific cellular components.
3. ***Phase-Contrast Microscopy*** – Enhances contrast in transparent, unstained specimens.
4. ***Confocal Microscopy*** – Uses lasers to produce high-resolution, three-dimensional images.

4. Write three apparatus required for macroscopic analysis of urine.

Three apparatus commonly required for macroscopic analysis of urine are:

1. ***Test tube*** – For collecting and examining urine samples.
2. ***Measuring cylinder*** – For measuring urine volume.
3. ***Urine specimen container*** – For collecting and storing the urine sample.

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5. Differentiate between polar and non-polar lipids.

Difference Between Polar and Non-Polar Lipids

Polar Lipids	Non-Polar Lipids
Contain both hydrophilic (polar) and hydrophobic (non-polar) regions.	Consist mainly of hydrophobic (non-polar) regions.
Soluble in polar and some organic solvents.	Insoluble in water; soluble in non-polar organic solvents.
Major components of biological membranes.	Mainly used for energy storage.
Examples: Phospholipids, glycolipids.	Examples: Triglycerides, waxes.
Possess charged or polar functional groups.	Lack charged or polar functional groups.

6. How glycogen is stored in liver?

Glycogen is stored in the liver as cytoplasmic glycogen granules within liver cells (hepatocytes). It is a highly branched polysaccharide composed of glucose units linked by α(1→4) glycosidic bonds with α(1→6) branch points. These granules serve as an energy reserve and can be rapidly broken down to release glucose when blood sugar levels fall, helping maintain normal blood glucose concentration.

7. Write effect of high temperature and solvent concentration on color of Iodine during starch test.

High temperature: The blue-black color disappears or fades because heating disrupts the amylose helix that traps iodine molecules. The color returns on cooling.

Solvent/concentration effect: A higher starch concentration produces a more intense blue-black color, while dilution results in a lighter or less distinct color due to fewer starch–iodine complexes being formed.

8. What is the turbidity of urine? What are the potential causes of turbidity of urine?

Normally, fresh urine is clear to slightly transparent, while increased turbidity may indicate infection or other urinary abnormalities. Turbidity of urine refers to the degree of cloudiness or haziness of urine caused by the presence of suspended particles.

Potential causes of urine turbidity include:

- Presence of pus cells (white blood cells)
- Bacteria due to urinary tract infection (UTI)
- Red blood cells
- Epithelial cells
- Crystals (urates, phosphates, oxalates)
- Mucus threads
- Yeast or fungal cells
- Semen contamination
- Excess protein or lipid content

9. Write role of yeast in decomposition of ripe fruits.

Yeast plays an important role in the decomposition of ripe fruits by fermenting the sugars present in the fruit. It converts glucose and other sugars into ethanol (alcohol) and carbon dioxide through the process of fermentation. This breakdown contributes to fruit spoilage, softening, and the characteristic fermented odor of overripe or decaying fruits. Yeast also helps recycle organic matter by initiating the decomposition process.

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10. Write principle of Fehling's test.

Fehling's test is based on the principle that reducing sugars reduce the blue-colored cupric ions (Cu^{2+}) of Fehling's solution to red or brick-red cuprous oxide (Cu_2O) in an alkaline medium upon heating. The formation of a brick-red precipitate indicates the presence of reducing sugars such as glucose, fructose, lactose, and maltose.

11. What are key recommendations for patient before collecting urine sample?

- ✓ Use a clean, dry, and sterile container for urine collection.
- ✓ Wash hands and clean the genital area before collecting the sample.
- ✓ Collect a midstream urine sample to minimize contamination.
- ✓ Avoid touching the inside of the container or lid.
- ✓ Preferably collect the first morning urine if instructed.
- ✓ Avoid excessive fluid intake before sample collection unless directed otherwise.
- ✓ Inform the healthcare provider about any medications being taken, as some may affect test results.
- ✓ Deliver the sample to the laboratory as soon as possible, ideally within 1–2 hours of collection.

12. Write main steps of lipids extraction.

Main steps of lipid extraction:

1. ***Sample preparation:*** The sample is dried and finely ground to remove moisture and increase the surface area for efficient extraction.
2. ***Addition of solvent:*** A non-polar organic solvent such as hexane, petroleum ether, chloroform, or diethyl ether is added because lipids are readily soluble in these solvents.
3. ***Extraction of lipids:*** The sample is mixed with the solvent and allowed to stand or heated gently (e.g., in a Soxhlet apparatus). During this process, lipids dissolve into the solvent.
4. ***Separation of extract:*** The solvent containing dissolved lipids is separated from the solid residue by filtration or decantation.
5. ***Evaporation of solvent:*** The solvent is removed by evaporation, leaving behind the extracted lipids.
6. ***Collection of lipids:*** The recovered lipids are collected, weighed, and analyzed if required to determine lipid content.

13. Differentiate between unsaturated, saturated and super saturated solutions.

Unsaturated Solution	Saturated Solution	Supersaturated Solution
Contains less solute than the maximum amount that can dissolve at a given temperature.	Contains the maximum amount of solute that can dissolve at a given temperature.	Contains more dissolved solute than a saturated solution at the same temperature.
More solute can still be dissolved.	No more solute can dissolve under normal conditions.	Additional solute cannot remain dissolved and may crystallize easily.
Stable solution.	Stable solution in equilibrium with undissolved solute.	Unstable solution.
No undissolved solute is present.	Undissolved solute may be present at the bottom.	Usually prepared by heating and then carefully cooling a saturated solution.
Example: A sugar solution in which more sugar can still dissolve.	Example: A sugar solution that cannot dissolve any more sugar.	Example: A cooled sugar syrup containing more dissolved sugar than normally possible at that temperature.

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14. What is the purpose of solvent saturation in TLC plate?

Solvent saturation in TLC (Thin Layer Chromatography) is done to fill the developing chamber with solvent vapors before the plate is developed**.** This minimizes solvent evaporation from the TLC plate, ensures uniform movement of the solvent front, improves separation of compounds, and produces more accurate and reproducible Rf values. Solvent saturation also helps prevent distortion or tailing of spots during chromatographic separation.

15. What is lyophilization? Write its applications.

Lyophilization (freeze-drying) is a dehydration process in which a material is first frozen and then the ice is removed directly as vapor by sublimation under reduced pressure (vacuum)**.** This method preserves the structure and biological activity of heat-sensitive substances.

Applications:

- Preservation of microorganisms and cell cultures.
- Storage of vaccines, antibiotics, and other pharmaceutical products.
- Drying of enzymes, proteins, and biological samples.
- Preservation of food products such as coffee, fruits, and dairy products.
- Long-term storage of blood plasma and other medical materials.
- Preparation of stable laboratory and research reagents.

16. Why the solution is boiled after adding the Fehling's reagent? What is the end product?

The solution is boiled after adding Fehling's reagent to provide the energy required for the redox reaction between the reducing sugar and the copper ions in Fehling's solution. Heating accelerates the reaction and allows the reducing sugar to reduce cupric ions (Cu^{2+}) to cuprous ions (Cu^{+}).

The end product is a brick-red precipitate of cuprous oxide (Cu_2O), which indicates the presence of a reducing sugar. The sugar itself is oxidized to the corresponding carboxylic acid (or its salt in the alkaline medium).

17. Write principle of Iodine test.

The iodine test is based on the principle that iodine molecules form a complex with the helical structure of amylose, a component of starch. When iodine solution is added to starch, the iodine molecules become trapped within the amylose helix, producing a characteristic blue-black color. The appearance of this color indicates the presence of starch.

18. Define Molarity and Molality.

Molarity (M):

Molarity is the concentration of a solution expressed as the number of moles of solute dissolved in one liter (1 dm^3) of solution. It is written as:

$$M = \frac{moles\ of\ solute}{volume\ of\ solution\ in\ litres}$$

Since it depends on the volume of solution, molarity changes with temperature because volume expands or contracts with temperature changes.

Molality (m):

Molality is the concentration of a solution expressed as the number of moles of solute present in one kilogram of solvent. It is written as:

$$m = \frac{moles\ of\ solute}{mass\ of\ solvent\ in\ kg}$$

It does not change with temperature because it is based on mass, which remains constant.

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19. Name different isomers of glucose.

Glucose shows different types of isomerism:

1. ***D- and L-glucose (Optical isomers / enantiomers):*** These are mirror images of each other based on the configuration of the chiral carbon farthest from the carbonyl group.
2. ***α-D-glucose and β-D-glucose (Anomers):*** These differ in the orientation of the hydroxyl group at the anomeric carbon (C-1 in cyclic form). They are formed when glucose cyclizes.
3. ***Open-chain and cyclic forms (Structural isomers within same compound):*** Glucose exists in equilibrium between a straight-chain aldehyde form and cyclic hemiacetal forms (pyranose and furanose forms).

20. Define polysaccharides and its types.

Polysaccharides are complex carbohydrates composed of a large number of monosaccharide units (usually glucose) joined together by glycosidic bonds. They serve mainly as storage or structural carbohydrates in living organisms.

Types of polysaccharides:

1. ***Homopolysaccharides:***
   These are made up of only one type of monosaccharide unit.
   Examples:
   - Starch (storage form in plants)
   - Glycogen (storage form in animals)
   - Cellulose (structural component in plants)
2. ***Heteropolysaccharides:***
   These are made up of two or more different types of monosaccharide units.
   Examples:
   - Hyaluronic acid
   - Heparin
   - Chondroitin sulfate

21. What color amylose and amylopectin give in the presence of starch?

Amylose gives a deep blue to blue-black color in the presence of iodine because its long, unbranched helical structure traps iodine molecules and forms a stable complex.

Amylopectin gives a reddish-brown or purple color because its highly branched structure does not form a strong iodine complex like amylose, resulting in a weaker and different color reaction.

LONG QUESTIONS

1. Write procedure for preparation of 5% NaCl solution.

Procedure for preparation of 5% NaCl solution:

1. ***Calculate required amount:*** A 5% (w/v) NaCl solution means 5 g of sodium chloride is present in 100 mL of solution. So, weigh 5 g of NaCl using an analytical balance.
2. ***Dissolve solute:*** Transfer the weighed NaCl into a clean beaker and add about 60–70 mL of distilled water. Stir using a glass rod until the salt is completely dissolved.
3. ***Transfer to volumetric flask:*** Pour the solution into a 100 mL volumetric flask using a funnel. Rinse the beaker and funnel with distilled water and add the washings to the flask to ensure complete transfer.
4. ***Make up the volume:*** Add distilled water carefully until the solution reaches the 100 mL mark on the volumetric flask.
5. ***Mix properly:*** Close the flask and invert it several times to ensure uniform mixing of the solution.

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2. How glycogen is stored as energy? Explain.

Glycogen is stored as an energy reserve mainly in the liver and muscles of animals. It is a highly branched polysaccharide of glucose, which allows it to be rapidly broken down when energy is needed.

After food intake, excess glucose in the blood is converted into glycogen through glycogenesis under the influence of insulin. In this process, glucose molecules are linked together by α(1→4) glycosidic bonds with α(1→6) branch points, forming compact glycogen granules stored in the cytoplasm of hepatocytes (liver cells) and muscle cells.

When the body requires energy, glycogen is broken down by glycogenolysis into glucose-1-phosphate and then glucose-6-phosphate. In the liver, this can be converted into free glucose and released into the bloodstream to maintain blood glucose levels. In muscles, it is directly used to produce ATP for muscular activity.

Thus, glycogen acts as a readily mobilizable energy source that helps maintain energy balance in the body.

3. Write complete procedure of TLC plate.

Procedure for preparation and use of a TLC (Thin Layer Chromatography) plate:

1. ***Preparation of adsorbent slurry:*** A suitable adsorbent such as silica gel or alumina is mixed with distilled water or an appropriate solvent to form a smooth slurry without lumps.
2. ***Coating of plate:*** A clean glass, aluminum, or plastic plate is uniformly coated with the prepared slurry using a spreader to form a thin, even layer.
3. ***Drying of plate:*** The coated plate is allowed to air dry and then activated by heating in an oven (usually at 100–120°C) to remove moisture and improve adsorption properties.
4. ***Spotting of sample:*** A small amount of the sample solution is applied as a spot near the base line of the TLC plate using a capillary tube.
5. ***Development of chromatogram:*** The plate is placed in a developing chamber containing a suitable solvent system. The solvent level is kept below the sample spot. The chamber is closed to allow solvent saturation.
6. ***Solvent movement:*** The solvent rises through capillary action, carrying components of the sample at different rates depending on their affinity for the stationary phase (adsorbent) and mobile phase (solvent).
7. ***Drying and visualization:*** After the solvent front reaches a marked height, the plate is removed, dried, and spots are visualized under UV light or by using a suitable detecting agent.
8. ***Calculation (if required):*** The Rf value of each component is calculated using:

$$Rf = \frac{Distance\ traveled\ by\ solute}{Distance\ traveled\ by\ solvent\ front}$$

4. Write process of fermentation by yeast.

Process of fermentation by yeast:

Fermentation by yeast is an anaerobic process in which yeast (mainly *Saccharomyces cerevisiae*) converts sugars such as glucose into ethanol and carbon dioxide in the absence of oxygen.

First, glucose enters the yeast cell and undergoes glycolysis, where it is broken down into two molecules of pyruvate, producing a small amount of energy (ATP) and NADH. Since oxygen is absent, pyruvate cannot enter aerobic respiration.

Instead, pyruvate is converted into acetaldehyde by releasing carbon dioxide. The enzyme pyruvate decarboxylase catalyzes this step. Then, acetaldehyde is reduced to ethanol by alcohol dehydrogenase, using NADH and regenerating NAD^+, which is essential to continue glycolysis. The overall reaction is:

Glucose → Ethanol + Carbon dioxide + Energy (ATP)

This process is widely used in baking (for bread rising) and in alcoholic beverage production.

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5. Write procedure for Fehling's test of urine.

Procedure for Fehling's test of urine (detailed):

1. ***Preparation of reagents:*** Fehling's test uses two separate solutions—Fehling's A (copper sulfate solution) and Fehling's B (alkaline solution of sodium potassium tartrate in sodium hydroxide). These are always mixed freshly before use because the active reagent is unstable.
2. ***Mixing of reagents:*** Take a clean, dry test tube and add equal volumes (about 1 mL each) of Fehling's A and Fehling's B. Mix them well. The solution should appear deep blue due to the presence of cupric ions in alkaline medium.
3. ***Addition of urine sample:*** Add about 2–3 mL of the urine sample to the freshly prepared Fehling's solution. Mix gently to ensure proper contact between the reagents and sample.
4. ***Heating:*** Place the test tube in a boiling water bath or heat it gently over a flame for about 5–10 minutes. Direct heating should be avoided to prevent decomposition or false results.
5. ***Observation:*** Observe the change in color and formation of precipitate during heating.

Result interpretation:

If reducing sugars (such as glucose) are present in urine, the blue cupric ions (Cu^{2+}) are reduced to cuprous oxide (Cu_2O), forming a brick-red precipitate. If no reducing sugar is present, the solution remains blue with no precipitate.

6. Write significance of color, clarity and pH of urine.

Significance of color, clarity, and pH of urine:

1. Color of urine:

The color of urine indicates the hydration status and possible presence of disease. Normal urine is pale yellow due to the pigment urochrome. Dark yellow urine suggests dehydration, while very pale urine may indicate overhydration. Abnormal colors such as red or brown may indicate the presence of blood, hemoglobin, bile pigments, or certain drugs and diseases.

2. Clarity (turbidity) of urine:

Clarity refers to how transparent or cloudy urine is. Normal urine is usually clear. Cloudy or turbid urine may indicate the presence of suspended particles such as bacteria, pus cells, epithelial cells, crystals, or mucus. It can be a sign of urinary tract infection or other kidney disorders.

3. pH of urine:

Urine pH reflects the acid–base balance of the body and diet. Normal urine pH ranges from about 4.5 to 8. Acidic urine may be due to high protein diet, starvation, or metabolic acidosis, while alkaline urine may indicate urinary tract infection or a vegetarian diet. Monitoring pH helps in diagnosing metabolic and renal disorders.

7. Why amylose appear blue-black with iodine?

Amylose appears blue-black with iodine due to the formation of a specific inclusion complex between iodine and the helical structure of amylose.

Amylose is a linear polysaccharide made up of glucose units linked by α(1→4) glycosidic bonds. Because of this structure, it naturally forms a coiled or helical shape in solution. When iodine solution (I_2 in KI) is added, iodine molecules enter into the central cavity of this amylose helix.

Inside the helix, iodine reacts with iodide ions to form polyiodide chains (I_3^-, I_5^-), which get trapped within the amylose structure. This inclusion complex alters the way light is absorbed and reflected, producing a deep blue-black coloration.

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The intensity of this color depends on the amount and integrity of amylose helices present. When the helix structure is disrupted (for example, by heating), the complex breaks and the color disappears, which is why it is reversible on cooling.

8. Differentiate between Starch and Glycogen.

Difference between Starch and Glycogen

Starch	Glycogen
Starch is the storage polysaccharide in plants.	Glycogen is the storage polysaccharide in animals and fungi.
It is composed of amylose (linear) and amylopectin (branched).	It is highly branched polysaccharide only.
Branching is less frequent (amylopectin has fewer branches).	Branching is more frequent and compact.
Stored mainly in seeds, roots, and tubers.	Stored mainly in liver and muscles.
Gives blue-black (amylose) or reddish-brown (amylopectin) color with iodine.	Gives reddish-brown color with iodine.
Less soluble in water.	More soluble than starch due to extensive branching.
Used for long-term energy storage in plants.	Used for short-term energy storage and rapid glucose release in animals.

9. What is Retention factor (Rf) in chromatography? Write its formula.

Retention factor (Rf) in chromatography is a numerical value that represents how far a substance moves on a chromatographic medium (such as a TLC plate) relative to the movement of the solvent front. It helps in identifying and comparing different compounds based on their mobility under specific conditions.

The Rf value is calculated using the formula:

Rf = Distance traveled by the solute (spot) ÷ Distance traveled by the solvent front

The value of Rf is always between 0 and 1. A higher Rf value indicates that the compound has a greater affinity for the mobile phase, while a lower Rf value indicates stronger interaction with the stationary phase.

10. Explain extraction of lipids by tissue sample.

Extraction of lipids from a tissue sample:

- ✓ Lipids are extracted from tissue samples using organic solvents because they are non-polar and soluble in non-polar solvents.
- ✓ First, the tissue sample is collected, cleaned, and finely homogenized to break the cells and increase surface area for extraction. The homogenized tissue is then dried if necessary to remove moisture, as water can interfere with lipid extraction.
- ✓ A suitable organic solvent such as a mixture of chloroform and methanol (as in the Folch method) or hexane is added to the tissue. The mixture is thoroughly mixed or shaken so that lipids dissolve into the solvent phase.
- ✓ After mixing, the solution is allowed to stand or is centrifuged to separate layers. Typically, two phases form: a lower organic phase containing lipids and an upper aqueous phase containing non-lipid impurities.
- ✓ The lipid-containing organic layer is carefully collected. Finally, the solvent is removed by evaporation, leaving behind the extracted lipids, which can then be weighed or analyzed further.

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